

Revised Generation Interconnection System Impact Study Report

For Queue Project Q14-0418

Bearskin 138 kV
May 27, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0418. The Interconnection Customer proposes a 338.7 MW wind facility interconnecting at the Bearskin 138 kV substation in the Virginia area. The request was received on December 27, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 226 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$57,049,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Lyondell - Chatham 138 KV L	Riverview - Comox 138 KV line	200.5	298	100.0
Bearskin - Chatham 138 KV	Riverview - Comox 138 KV	196.5	308	100.0
Comox - Riverview 138 KV L	Chatham - Bearskin 138 KV line	177.5	342	100.0
Lyondell - Leesville Substation 138 KV	Riverview - Comox 138 KV	164.2	198	75.4
Dixleta - N Pole 138 KV line	Dixleta - North Pole 138 KV Cir. 2	128.8	465	5.2
North Pole - Dixleta 138 KV Cir. 2	Dixleta - N Pole 138 KV Ckt 1	126.7	472	5.2
Hughson - Dileta 345 KV	Rustic - Dixleta 345 KV	125.6	1,673	17.5
Westport - Wheelabrator Baltimore Refuse 138 KV	Dixleta - Taber 345 KV	123.9	589	5.1
N Pole - High Ridge 138 KV	Taber - Glenannan 345 KV	123.4	799	7.4
Dixleta - Rustic 345 KV	Hughson - Dixleta 345 KV	122.5	1,623	15.9
Jerome - High Ridge 138 KV Line	Taber Substation - Glenannan 345 KV	117.9	868	7.4
Sunbright - Failand 138 KV	Glenannan - Taber 345 KV	117.4	877	6.4
Sunbright - Colesville 138 KV	Taber - Glenannan 345 KV	115.3	941	6.4
Dixleta - Drum 138 KV	Taber - Glenannan 345 KV	113.7	818	5.3
Taber - Glenannan 345 KV	Sierra Pacific Lumber Substation 138 KV/1 kV Autotransformer	112.9	2,382	17.4
Taber - Dixleta 345 KV	Sierra Pacific Lumber 138-1 kV Transformer	111.6	2,258	15.7

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Rustic - Hughson 345 KV	Dixleta - Rustic 345 KV	111.3	1,540	12.0
Aibonito Pds - Shenandoah 138 KV Line	Clover - Bear 345 KV	107.0	238	5.7
Lyondell - Orlando Cogen Lp 138 KV	Riverview - Comox 138 KV	106.8	243	24.6
Riva Rd - Sierra Pacific Lumber 345 KV	Taber - Glenannan 345 KV L	106.6	1,220	6.1
Wlkens - Highgrove 138 KV	Dixleta - Taber 345 KV	106.4	649	5.1
Louisa Generation Facility - North Anna 345 KV L	Louisa Generation Facility - Carrabelle 345 KV Line	105.4	2,070	13.2
Wilkens - Wheelabrator Baltimore Refuse 138 KV	Taber - Dixleta 345 KV	105.1	694	5.1
Sierra Pacific Lumber 345-1 kV TX	Taber - Glenannan 345 KV	105.0	1,239	6.1
Sierra Pacific Lumber 138 kV/1 kV Transformer	Taber - Glenannan 345 KV	105.0	1,239	6.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Chatham - Lyondell 138 KV	Rebuild line with higher-capacity conductor	\$14,844,000	68% (\$10,064,000)
Chatham - Bearskin 138 KV	Rebuild line with higher-capacity conductor	\$5,690,000	68% (\$3,857,000)
Riverview - Comox 138 KV	Rebuild line with higher-capacity conductor	\$2,109,000	68% (\$1,430,000)
Lyondell - Leesville Substation 138 KV	Rebuild line with higher-capacity conductor	\$14,658,000	68% (\$9,938,000)
North Pole - Dixleta 138 KV #1	Rebuild line with higher-capacity conductor	\$960,000	9% (\$86,000)
Dixleta - N Pole 138 KV Circuit 2	Rebuild line with higher-capacity conductor	\$586,000	9% (\$53,000)
Hughson - Dileta 345 KV	Rebuild line with higher-capacity conductor	\$42,290,000	7% (\$2,894,000)
Westport - Wheelabrator Baltimore Refuse 138 KV	Rebuild line with higher-capacity conductor	\$841,000	11% (\$95,000)
High Ridge - N Pole 138 KV	Rebuild line with higher-capacity conductor	\$3,020,000	8% (\$235,000)
Rustic - Dixleta 345 KV	Rebuild line with higher-capacity conductor	\$87,386,000	7% (\$5,957,000)
Jerome - High Ridge 138 KV Line	Rebuild line with higher-capacity conductor	\$2,306,000	8% (\$180,000)
Sunbright - Failand 138 KV	Rebuild line with higher-capacity conductor	\$606,000	8% (\$47,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Sunbright - Colesville 138 KV	Rebuild line with higher-capacity conductor	\$1,235,000	8% (\$96,000)
Dixleta - Drum 138 KV	Rebuild line with higher-capacity conductor	\$1,937,000	8% (\$158,000)
Taber - Glenannan 345 KV	Rebuild line with higher-capacity conductor	\$39,358,000	7% (\$2,729,000)
Taber - Dixleta 345 KV	Rebuild line with higher-capacity conductor	\$28,997,000	7% (\$2,009,000)
Rustic - Hughson 345 KV	Rebuild line with higher-capacity conductor	\$46,852,000	7% (\$3,144,000)
Aibonito Pds - Shenandoah 138 KV Line	Reconductor line	\$10,622,000	47% (\$5,001,000)
Lyondell - Orlando Cogen Lp 138 KV	Reconductor line	\$5,092,000	68% (\$3,451,000)
Riva Rd - Sierra Pacific Lumber 345 KV	Reconductor line	\$18,748,000	8% (\$1,497,000)
Wlkens - Highgrove 138 KV	Reconductor line	\$230,000	12% (\$28,000)
Louisa Generation Facility - N Anna Substation 345 KV	Reconductor line	\$50,662,000	6% (\$2,887,000)
Wilkens - Wheelabrator Baltimore Refuse 138 KV	Reconductor line	\$461,000	12% (\$57,000)
Sierra Pacific Lumber 345-1 kV TX	Replace transformer with higher-rated unit	\$9,812,000	8% (\$783,000)
Sierra Pacific Lumber 138 kV/1 kV Transformer	Replace transformer with higher-rated unit	\$4,669,000	8% (\$373,000)
	Total allocated to project		\$57,049,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.